

instead fit their brains to their actions. These go back and cheat with the statistics to justify their actions. In my business, they fool themselves with statistical arguments to justify their option selling. -

### *Probabilities and the Media (More Journalists)*

A journalist is trained in methods to express himself rather than to plumb the depth of things – the selection process favors the most communicative, not necessarily the most knowledgeable. My medical doctor friends claim that many medical journalists do not understand anything about medicine and biology, often making mistakes of a very basic nature. I cannot confirm such statements, being myself a mere amateur (though at times a voracious reader) in medical research but I have noticed that they almost always misunderstand the probabilities used in medical research announcements. The most common one concerns the interpretation of evidence. They most commonly get mixed up between *absence of evidence* and *evidence of absence*, a similar problem to the one we saw in Chapter 9. How? Say I test some chemotherapy, say Fluorouracil for upper respiratory tract cancer, and find that it is better than a placebo, but only marginally so; that (in addition to other modalities) it improves survival from 21 per 100 to 24 per 100. Given my sample size, I may not be confident that the additional 3% survival points come from the medicine; it could be merely attributable to randomness. I would write a paper outlining my results and saying that there is no evidence of improved survival (as yet) from such medicine, and that further research would be needed. A medical journalist would pick it up and claim that one Professor N.N. Taleb found evidence that Fluorouracil *does not help*, which is entirely opposite to my intentions. Some naïve doctor in Smalltown, even more uncomfortable with probabilities than the most untrained journalist, would pick it up and build a mental block against the medication, even when some researcher finally finds fresh evidence that such medicine confers a clear survival advantage.

### CNBC AT LUNCH TIME

The advent of the financial television channel CNBC presented plenty of benefits to the financial community but it also allowed a collection of extrovert practitioners long on theories to voice them in a few minutes of television time. One often sees respectable people making ludicrous (but smart-sounding) statements about properties of the stock market. Among these are statements that blatantly violate the laws of probability. One summer during which I was assiduous at the health club, I often heard such statement as “the real market is only 10% off the highs while the average stock is close to 40% off its highs”, which is intended to be indicative of deep troubles or anomalies – some harbinger of bear markets.

There is no incompatibility between the fact that the average stock is down 40% from the highs while the average of all stocks (that is, the market) is down 10% from its own highs. One must consider that the stocks did not all reach their highs *at the same time*. Given that stocks are not 100% correlated, stock A might reach its maximum in January, stock B might reach its maximum in April, but the average of the two stocks A and B might reach its maximum at some time in February. Furthermore, in the event of negatively correlated stocks, if stock A is at its maximum when stock B is at its minimum, then they could both be down 40% from their maximum when the stock market is at its highs! By a law of probability called distribution of the maximum of random variables, the maximum of an average is necessarily less volatile than the average maximum.

### YOU SHOULD BE DEAD BY NOW

This brings to mind another common violation of probability by prime-time T.V. financial experts, who may be selected for their looks, charisma and their presentation skills, but certainly not for their incisive minds. For instance, a fallacy that I saw commonly made by a prominent T.V. financial guru goes as follows: “The average American is expected to live 73 years. Therefore if you are 68 you can expect to live five more years, and should plan accordingly”. She went into precise prescriptions of how the person should invest for a five more years